



Liver Tumors, Function Tests & Failure

Pathology GIT Block · MEDucation by Mattin Majid · Practice MCQs

HOW TO USE THIS SET

20 questions, mixed difficulty, 5 case-based. Attempt every question before checking the answer key on the last page.

● 4 Easy

● 10 Medium

● 6 Hard

1.**EASY**

Which enzymes reflect hepatocyte integrity and rise with hepatocyte destruction?

- A. Serum ammonia, bile acids
- B. AST, ALT, LDH
- C. AFP, CEA
- D. Albumin, PT, PTT
- E. Alkaline phosphatase, GGT

2.**EASY**

Acute liver failure is defined as encephalopathy developing within how many months of initial diagnosis?

- A. 12 months
- B. 3 months
- C. 1 month
- D. 24 months
- E. 6 months

3.**EASY**

Which is the most common benign tumor of the liver?

- A. Focal nodular hyperplasia
- B. Hepatoblastoma
- C. Hepatic hemangioma
- D. Hepatocellular adenoma
- E. Bile duct adenoma

4.**EASY**

Which is the commonest primary malignant tumor of the liver in adults?

- A. Hepatoblastoma
- B. Angiosarcoma
- C. Hepatocellular carcinoma
- D. Metastatic carcinoma
- E. Cholangiocarcinoma

5.**MEDIUM**

In the US, which cause accounts for almost 50% of acute liver failure cases?

- A. Hepatitis B
- B. Hepatitis E
- C. Alcohol
- D. Autoimmune hepatitis
- E. Acetaminophen ingestion

6.**MEDIUM**

In Asia, which viruses predominate as causes of acute liver failure?

- A. Hepatitis A and B
- B. CMV and EBV
- C. Hepatitis C and D
- D. Hepatitis B and E
- E. Hepatitis A and C

7.

MEDIUM

Which neurologic sign is typical of hepatic encephalopathy?

- A. Myoclonus of the face
- B. Babinski sign
- C. Pill-rolling tremor
- D. Nystagmus
- E. Asterixis

8.

MEDIUM

In the HEPATICS mnemonic for precipitants of hepatic encephalopathy, what does the "A" represent?

- A. Anemia
- B. Alcohol withdrawal
- C. Alkalosis, azotemia
- D. Antibiotics
- E. Ascites

9.

MEDIUM

What percentage of individuals with cirrhosis are asymptomatic?

- A. 10%
- B. 40%
- C. 70%
- D. 90%
- E. 100%

10.

MEDIUM

CASE

A cirrhotic patient develops dilated veins around the umbilicus (caput medusae) and esophageal varices.

These findings are a direct consequence of:

- A. Hyperestrogenemia
- B. Portal hypertension with portosystemic shunting
- C. Hypersplenism
- D. Hepatic encephalopathy
- E. Coagulopathy

11.

MEDIUM

Hepatocellular adenoma is most strongly associated with which risk factor?

- A. Alcohol use
- B. Aflatoxin exposure
- C. Hemochromatosis
- D. Chronic hepatitis B infection
- E. Oral contraceptive use

12.

MEDIUM

Hepatic hemangiomas larger than approximately what size are liable for hemorrhage and consumptive coagulopathy?

- A. 1 cm
- B. 2 cm
- C. 10 cm
- D. 5 cm
- E. 15 cm

13.

MEDIUM

Which plasma membrane enzymes rise with damage to the bile canaliculus, reflecting biliary excretory dysfunction?

- A. Alkaline phosphatase and GGT
- B. Amylase and lipase
- C. CK and troponin
- D. AST and ALT
- E. LDH and CRP

14.

MEDIUM

Which test reflects hepatocyte synthetic function by assessing coagulation?

- A. Prothrombin time (PT)
- B. Serum AST
- C. Serum bile acids
- D. Urine bilirubin
- E. Serum alkaline phosphatase

15.

HARD CASE

A patient with cirrhosis develops a rapid deterioration in liver function after an episode of severe gastrointestinal bleeding, superimposed on years of stable, compensated chronic liver disease.

This presentation is best classified as:

- A. Acute liver failure
- B. Chronic liver failure alone
- C. Acute-on-chronic liver failure
- D. Hepatorenal syndrome
- E. Simple decompensated cirrhosis

16.

HARD

Which combination of gross and microscopic findings best defines cirrhosis?

- A. Multiple well-circumscribed pale nodules in an otherwise normal liver
- B. Massive necrosis with minimal nodularity and no fibrosis
- C. Diffuse hepatocyte swelling without nodularity or fibrosis
- D. Regenerative parenchymal nodules surrounded by fibrous bands, with loss of the normal central-portal relationship
- E. A single large well-circumscribed mass with normal surrounding architecture

17.

HARD CASE

A patient from a region of sub-Saharan Africa with chronic hepatitis B acquired via vertical transmission in infancy develops a large hepatic mass with mildly elevated AFP.

Which statement best reflects the role of AFP in this scenario?

- A. AFP is elevated in only about half of advanced HCC cases and is neither sensitive nor specific
- B. AFP is diagnostic for HCC when elevated above any detectable level
- C. A normal AFP excludes hepatocellular carcinoma
- D. AFP is more useful for metastatic tumors than for HCC
- E. AFP elevation confirms vertical HBV transmission

18.

HARD

Which inherited metabolic disorders are most strongly linked to hepatocellular carcinoma?

- A. Glycogen storage disease type I only
- B. Phenylketonuria and galactosemia
- C. Cystic fibrosis and Fabry disease
- D. Hereditary hemochromatosis and alpha-1-antitrypsin deficiency
- E. Gaucher disease and Niemann-Pick disease

19.

HARD CASE

A liver resection specimen shows multiple well-circumscribed, pale nodules scattered throughout an otherwise unremarkable liver, in a patient with a known history of colon cancer.

This gross pattern is most consistent with:

- A. Angiosarcoma
- B. Hepatocellular adenoma
- C. Metastatic carcinoma
- D. Cirrhosis
- E. Hepatocellular carcinoma, diffuse pattern

20.

HARD CASE

A patient who worked for decades in a vinyl chloride manufacturing plant develops a rapidly growing, hemorrhagic liver mass composed of malignant cells forming anastomosing vascular channels on biopsy.

Which other exposures share this tumor's known association, besides vinyl chloride?

- A. Aflatoxin and chronic hepatitis B
- B. HFE gene mutations and iron overload
- C. Oral contraceptives and obesity
- D. Arsenic, Thorotrast, and androgenic steroids
- E. Acetaminophen and chlorpromazine

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	B	AST, ALT, and LDH are cytosolic hepatocyte enzymes that elevate when hepatocytes are destroyed.
2	E	Acute liver failure is defined as acute liver illness producing hepatic encephalopathy within 6 months of initial diagnosis.
3	C	Hepatic hemangiomas are benign vascular tumors and the most common benign tumor of the liver.
4	C	Hepatocellular carcinoma is the commonest primary malignant tumor of the liver in adults.
5	E	Accidental or deliberate acetaminophen ingestion accounts for almost 50% of acute liver failure cases in the US.
6	D	In Asia, acute hepatitis B and E predominate as causes of acute liver failure.
7	E	Asterix (a flapping tremor) is the typical neurologic sign of hepatic encephalopathy.
8	C	In the HEPATICS mnemonic, "A" stands for alkalosis and azotemia as precipitants of hepatic encephalopathy.
9	B	About 40% of individuals with cirrhosis are asymptomatic.
10	B	Sustained portal hypertension causes portosystemic shunting and dilation of collateral vessels, producing esophagogastric varices and caput medusae.
11	E	Hepatocellular adenoma is associated with oral contraceptive drug use, obesity/metabolic syndrome, and certain genetic mutations (e.g., beta-catenin).
12	D	Hemangiomas larger than about 5 cm are liable for hemorrhage, thrombosis, and consumptive coagulopathy.
13	A	Serum alkaline phosphatase and gamma-glutamyl transpeptidase (GGT) are plasma membrane enzymes that rise with damage to the bile canaliculus, part of biliary excretory function testing.
14	A	Prothrombin time (PT) and partial thromboplastin time (PTT) assess coagulation factors synthesized by hepatocytes, reflecting synthetic function.
15	C	Acute-on-chronic liver failure is a sudden acute liver failure developing in a patient with years of stable, compensated chronic liver disease, usually following a superimposed insult (here, GI hemorrhage, a HEPATICS precipitant).
16	D	Cirrhosis is defined by diffuse transformation into regenerative parenchymal nodules separated by bridging fibrosis, with loss of the normal central-portal relationship — distinct from acute liver failure (minimal scarring) or metastases (well-circumscribed nodules in normal background).
17	A	Serum AFP is elevated in only about 50% of individuals with advanced HCC and is neither a sensitive nor a specific marker.
18	D	Hereditary hemochromatosis and alpha-1-antitrypsin deficiency are the inherited disorders most strongly linked to HCC, with Wilson disease a lesser contributor.
19	C	Metastatic tumors, the most common liver malignancy overall, classically present as multiple well-circumscribed masses, distinguishing them from the more infiltrative or trabecular patterns of primary liver tumors.
20	D	Angiosarcoma, the commonest primary liver sarcoma, is associated with exposure to androgenic steroids, arsenic, Thorotrast, and vinyl chloride.

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Recalled and source questions are treated as evidence of what gets asked, not as a source of correct answers — verify against your lecture notes.